ML-Driven PVT-Aware Leakage Estimation

Team 1 “Leaky Lima” · 16nm HP PTM · VDD_{nom} = 1.0 V · 3-bit carry-lookahead adder

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Summary. Four neural-network transistor models (NMOS/PMOS × ON/OFF) predict all four terminal currents of a 16 nm FET under process, voltage and temperature variation. Composed into logic gates, they reproduce 3,600 SPICE gate simulations with a **mean accuracy of 99.10 %** (median error 0.69 %, 95th percentile 2.60 %). Solving the internal node voltages with the same models instead of taking them from SPICE gives 99.12 %. On the 204-transistor CLA (25,600 SPICE simulations) the hierarchical sum asked for in the brief reaches 96.41 %. Its error is almost entirely due to the hierarchy itself (isolated gates with ideal 0/VDD inputs), not the ML: the same sum of *SPICE* gate leakages is off by -2.64 %. A levels-aware hierarchy, which feeds each gate the ML-predicted output voltage of its driver, removes this bias and reaches **99.21 % mean accuracy** on the CLA (median error 0.66 %), with no SPICE voltage used anywhere.

1 Setup

Item	Value
Model file	16nm_HP.pm (PTM, BSIM4 level 54)
Nominal VDD	1.0 V (sampled ±10 %: 0.90–1.10 V)
Channel length	L = 16 nm; W = 1–8 × L = 16–128 nm
Gate sizing	W_n = 2L = 32 nm, W_p = 4L = 64 nm (interior of the training range)
Test circuit	3-bit carry-lookahead adder: 42 library gates, 204 transistors, 7 inputs (128 vectors)
Simulator	ngspice-45.2 : Circuit level simulation program
Software	Python 3.14.8, scikit-learn 1.9.0, LightGBM 4.7.0, numpy 2.5.4, pandas 3.1.1
Platform	macOS-27.0.1-arm64-arm-64bit-Mach-O, 11 cores
SPICE options	`.options gmin=1e-18 abstol=1e-18 reltol=1e-6 vntol=1e-9` in every netlist

Leakage definition. Gate and circuit leakage are the total static power divided by VDD, $P_{static}/VDD = \Sigma V_k I_k/VDD$ over the supply *and* the input sources. The input sources matter because gate-tunnelling current into an input pin is real leakage that a driving gate must supply. The plain supply current I(VDD) is also recorded (`*_ivdd` columns) and gives the same conclusions. Terminal currents are positive *into* the device, so $I_d+I_g+I_s+I_b = 0$.

Accuracy metric. Mean accuracy = 100 % − mean |ML/SPICE − 1|. Medians, 95th percentiles and the worst case are reported alongside it.

2 Data generation

2.1 Sampling

For each of the four cases, 10,000 samples are drawn with every parameter independent. VDD is drawn first, uniform within ±10 % of nominal. V_d and V_s are uniform in [0, VDD], W is uniform in [1, 8]×L and T is uniform in [−40, 125] °C. The five process parameters (vth0, toxe, ndep, u0, xj) are each uniform within ±10 % of the model-card nominal, separately for NMOS and PMOS. V_g and V_b are fixed by the case (ON: V_g = VDD for NMOS / 0 for PMOS; OFF: the opposite; body at the device's own rail). All draws use fixed seeds (`SEED_DATA = 2026`), and these 10,000-row files are deliverable 1.

A uniform draw almost never puts a terminal on a rail: 3 in 10,000 samples are within 2 % of both rails. Inside a gate, however, the leaking transistor always has its source on a rail and its drain at the output (≈ 0 or VDD). We

therefore add a **rail-focused supplement** of 5,000 samples per case. In it, V_d and V_s are exactly 0 or VDD one third of the time and otherwise follow an arcsine law that is dense near both rails. In half of these samples V_g is moved up to 10 % of VDD off its ideal level, because a gate driven by another gate sees a logic 0 a few tens of mV above ground. A third set of 2,000 samples per case at the gate widths and rail voltages (the *deployment check*) is used only for validation and test. No gate or circuit simulation is ever used for training or model selection.

2.2 Netlists

The single-FET netlist (one per case in `netlists/single_*.cir`) has one transistor and a separate voltage source on every terminal, so all four terminal currents are read directly through their sources. For speed, one ngspice process runs a whole batch of samples, changing the process parameters with `altermod` and the width, voltages and temperature with `alter` between `op` analyses. A self-check re-simulates samples as fresh standalone netlists and confirms the batch loop reproduces them. Besides the terminal currents, the BSIM4 internal current components (I_{ds}, I_{gs}, I_{gd}, I_{gcs}, I_{gcd}, I_{gb}, GIDL/GISL, I_{sub}, junction currents) are recorded. They are regrouped into six physical **bipoles** (ds, gs, gd, gb, db, sb), which rebuild the four terminal currents exactly. Gate and circuit netlists are generated from the *same* transistor list that the ML side uses, so SPICE and ML cannot disagree about topology. Each starts from a logic-level `.nodeset`; the reason is given in §6.

2.3 Splits

Each 10,000-row dataset is split 70/15/15 (7,000 train / 1,500 validation / 1,500 test) with `SEED_SPLIT = 42`. The rail supplement is split the same way and its training part is added to the training data. The test parts are touched once, after model selection. Split indices are saved in `results/split_*.npz`.

2.4 Data check

Table 1. Kirchhoff check, $|I_d+I_g+I_s+I_b|$ for each case (10,000 samples).

Case	samples	median $\|\Sigma I\|$ (A)	max $\|\Sigma I\|$ (A)	median $\|\Sigma I\|/\max\|I\|$	p95 $\|\Sigma I\|/\max\|I\|$
NMOS ON	10000	1.76×10^{-14}	9.88×10^{-13}	9.78×10^{-10}	4.34×10^{-8}
NMOS OFF	10000	5.62×10^{-20}	4.27×10^{-15}	8.93×10^{-10}	3.71×10^{-9}
PMOS ON	10000	7.60×10^{-15}	9.86×10^{-13}	8.63×10^{-10}	1.49×10^{-8}
PMOS OFF	10000	9.81×10^{-17}	8.38×10^{-15}	2.09×10^{-4}	2.04×10^{-2}

KCL closes to 10^{-9} of the largest terminal current in the median for every case. The largest relative residuals (PMOS OFF, up to 14.5 %) occur only where every terminal current is around 10^{-15} A. That is the simulator's absolute tolerance (abstol = 10^{-18} A amplified by the solver's convergence check), not a physical violation: the absolute residual never exceeds 9.88×10^{-13} A.

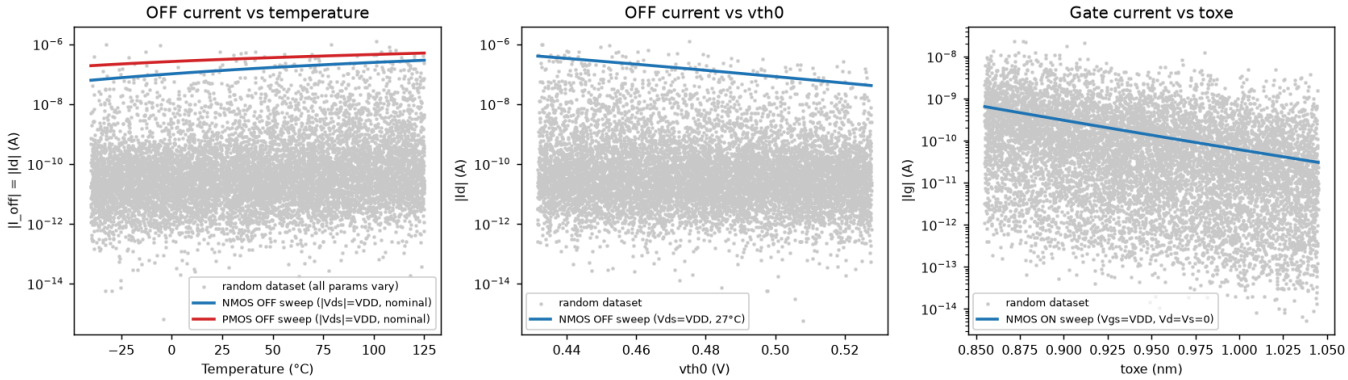


Figure 1. Trend checks. OFF current rises 2.19× from 25 to 125 °C (sub-threshold current $\propto \exp(-V_{th}/nV_T)$, and V_{th} falls with T). It falls to 0.31× with +10 % vth0 (exponential in V_{th}). Gate current falls to 0.23× with +10 % toxe (direct tunnelling is exponential in oxide thickness). All three trends match the physics.

3 Models

3.1 What is learned

Predicting the four terminal currents directly is hard: they span about 12 decades, change sign, and I_s is often the difference of two much larger opposite currents. We use the structure of the device instead. Each of the six bipoles is a passive two-terminal element whose current always flows from its higher to its lower terminal. The model therefore learns its **conductance** $G = I/V$, as log G per unit width, and rebuilds the terminal currents with KCL: $I_d = I_{ds} - I_{gd} + I_{db}$, and so on. This guarantees exact KCL, the correct sign of every current, and $I = 0$ at $V = 0$. Five sub-models per case share the inputs: channel (ds), gate-to-diffusion tunnelling (gs and gd, one model evaluated twice), gate-to-body, drain-junction and source-junction.

Features (per sub-model): VDD, T [K], $\log(W_{eff}/L)$, $\Delta V_g/V_T$ (offset of the gate from its ideal level), the five process parameters relative to nominal, the two terminal voltages of the bipole, V/V_T, $\log|V|$, and three Arrhenius-type inputs $1/V_T$, V_{th0}/V_T and $(VDD-V_{th0})/V_T$, since sub-threshold current goes as $\exp(-V_{th}/nV_T)$. The **width normalisation** follows BSIM4 for each mechanism. A width sweep in SPICE showed:

- The OFF channel scales with the *clamped* effective width. Below 20 nm BSIM4 replaces W_{eff} by $2{\cdot}10^{-8}(4{\cdot}10^{-8}-W_{eff})/(6{\cdot}10^{-8}-2W_{eff})$, so a 1L device conducts as if it were 14 nm wide, not 6 nm.
- The ON channel scales with the drawn W.
- Tunnelling scales with $W-2{\cdot}wint$.
- Junctions do not depend on W in this card.

Without the clamp the models carried a +2 % bias into every gate.

3.2 Model families and hyper-parameters

- **Polynomial + ridge**: standardised features, polynomial degree 3 or 4, ridge strength chosen by internal CV over 10^{-8}–1.
- **Random forest**: 300 trees, max_features 0.6, fully grown.
- **LightGBM**: 4,000 trees, learning rate 0.02, 31 or 63 leaves, row and column subsampling 0.8/0.9.
- **MLP**: 3 hidden layers of 64 or 128 tanh units, L-BFGS, L2 = 10^{-6}, up to 5,000 iterations, standardised inputs and target. Five networks with different seeds are averaged in log-G space (a geometric mean of the conductances), which removes most of L-BFGS's run-to-run noise.

All families use the same bipole targets and features. Selection uses the mean of the median-relative-error scores on three validation sets: uniform, rail-focused and deployment check. The table below is the brief's uniform validation set.

Table 2. Model comparison on the uniform validation set (1,500 samples per case). Errors are relative, on currents above 10^{-14} A; R^2 is computed on asinh(I/1 fA), which weights every decade equally. Bold = selected.

Case	Model	hyper-params	id R^2	id med %	id p95 %	ig R^2	ig med %	ig p95 %	is R^2	is med %	is p95 %	ib R^2	ib med %	ib p95 %
NMOS OFF	**MLP**	**3×64, 5 seeds**	**1.0000**	**0.18**	**2.30**	**1.0000**	**0.09**	**0.31**	**1.0000**	**0.18**	**2.07**	**0.9999**	**0.14**	**1.02**
NMOS OFF	LightGBM	leaves=31	0.9942	1.61	20.26	1.0000	0.75	2.27	0.9967	1.52	19.22	0.9982	0.57	10.02
NMOS OFF	Polynomial + ridge	degree=3	1.0000	0.13	1.77	1.0000	0.08	0.20	0.9910	0.15	2.06	0.7564	11.19	61.61
NMOS OFF	Random forest	trees=300	0.9867	2.06	49.92	0.9999	0.84	3.47	0.9912	2.23	48.49	0.9931	0.57	23.72
NMOS ON	**MLP**	**3×64, 5 seeds**	**1.0000**	**0.79**	**3.73**	**1.0000**	**0.25**	**1.17**	**1.0000**	**0.79**	**3.73**	**1.0000**	**0.39**	**1.46**
NMOS ON	LightGBM	leaves=31	1.0000	2.48	28.12	0.9995	3.71	12.09	1.0000	2.49	27.94	0.9987	6.73	24.70
NMOS ON	Polynomial + ridge	degree=4	0.9999	7.34	26.28	0.9958	10.34	33.03	1.0000	7.33	26.22	0.9981	8.93	25.31
NMOS ON	Random forest	trees=300	0.9998	6.07	56.88	0.9964	8.90	30.95	0.9999	6.07	56.26	0.9894	18.56	67.03
PMOS OFF	**MLP**	**3×64, 5 seeds**	**1.0000**	**0.28**	**2.09**	**0.9936**	**0.28**	**1.49**	**1.0000**	**0.51**	**2.33**	**0.9999**	**0.34**	**1.33**
PMOS OFF	LightGBM	leaves=31	0.9990	6.01	42.80	0.9935	1.17	3.69	0.9979	10.03	46.77	0.9962	1.07	12.40
PMOS OFF	Polynomial + ridge	degree=3	0.9998	1.79	18.36	0.9936	0.22	1.55	0.9977	1.26	25.22	0.9689	11.31	60.79
PMOS OFF	Random forest	trees=300	0.9946	17.34	139.94	0.9934	1.51	5.40	0.9954	28.46	171.13	0.9879	1.65	25.27
PMOS ON	**MLP**	**3×64, 5 seeds**	**1.0000**	**0.64**	**2.82**	**1.0000**	**0.43**	**1.53**	**1.0000**	**0.64**	**2.81**	**1.0000**	**0.39**	**1.33**
PMOS ON	LightGBM	leaves=31	1.0000	2.44	20.01	0.9991	7.23	22.43	1.0000	2.44	19.96	0.9991	6.36	20.70
PMOS ON	Polynomial + ridge	degree=4	1.0000	6.96	23.12	0.9969	8.96	27.97	0.9998	6.96	23.22	0.9982	8.98	27.36
PMOS ON	Random forest	trees=300	0.9999	6.62	40.84	0.9934	18.36	65.26	0.9999	6.62	40.81	0.9937	16.32	59.30

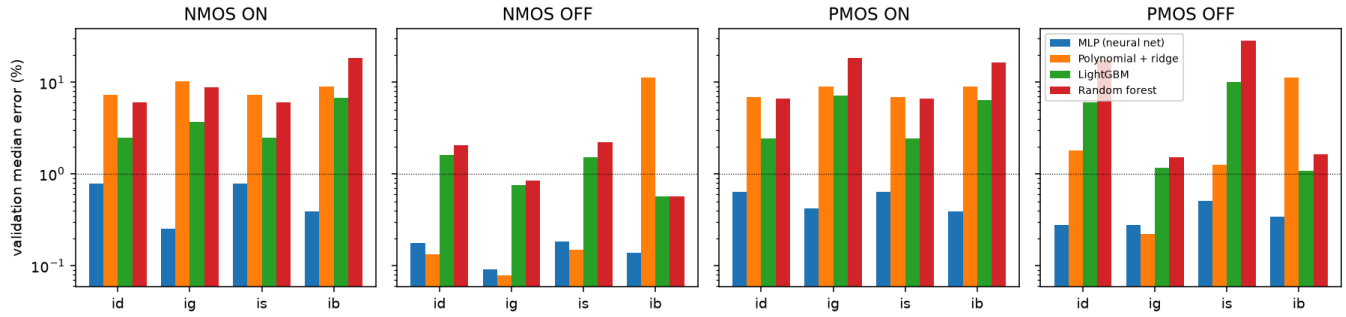


Figure 2. Validation median error by model family (log scale). The dotted line is 1 %.

Choice. The MLP wins every case, usually by 5–20× in median error and more in the 95th percentile. Log-conductance is a smooth function of the inputs, a sum of exponents with gentle non-linearities, which a tanh network represents with few parameters. Trees fit it in piecewise-constant steps, which gives the heavy p95 tails in Table 2. A degree-3/4 polynomial is accurate where it is flexible enough (NMOS OFF I_d) but fails on mechanisms with sharper bias dependence, for example the junction/GIDL current in I_b. Of the two MLP sizes, the 64-unit network was selected for every case.

3.3 Test-set performance of the chosen models

Table 3. Chosen models on the held-out test set (1,500 uniform samples per case).

Case	Model	id med %	ig med %	is med %	ib med %	id p95 %	ig p95 %	is p95 %	ib p95 %
NMOS ON	MLP	0.69	0.24	0.69	0.37	3.92	1.07	3.87	1.38
NMOS OFF	MLP	0.17	0.10	0.18	0.14	2.23	0.31	2.02	1.06
PMOS ON	MLP	0.59	0.41	0.59	0.38	2.90	1.60	2.90	1.39
PMOS OFF	MLP	0.32	0.30	0.53	0.33	2.00	1.70	2.26	1.45

Table 4. The same models on the deployment-check test set (1,000 samples per case at the gate widths and rail voltages).

Case	id med %	ig med %	is med %	ib med %	id p95 %	ig p95 %	is p95 %	ib p95 %
NMOS ON	1.60	0.73	1.59	0.38	7.25	1.77	7.16	1.25
NMOS OFF	0.83	0.12	0.83	2.57	2.66	0.47	2.71	15.23
PMOS ON	0.78	0.38	0.78	0.39	4.08	1.52	4.02	1.52
PMOS OFF	0.86	0.38	0.86	2.93	3.10	1.71	3.10	14.29

On the test set the median error is below 1 % for every current and case. On the deployment-check set the leakage-carrying currents (I_d, I_s) stay below 1 % in the median. The larger OFF-state I_b errors there are on body currents of about 10^{-14} A, six to seven decades below the channel current, so they do not affect gate leakage. The PMOS-OFF I_g and I_s rows evaluate fewer samples (see the `eval_frac` columns of `chosen_models_test.csv`), because those currents are often below the 10^{-14} A floor.

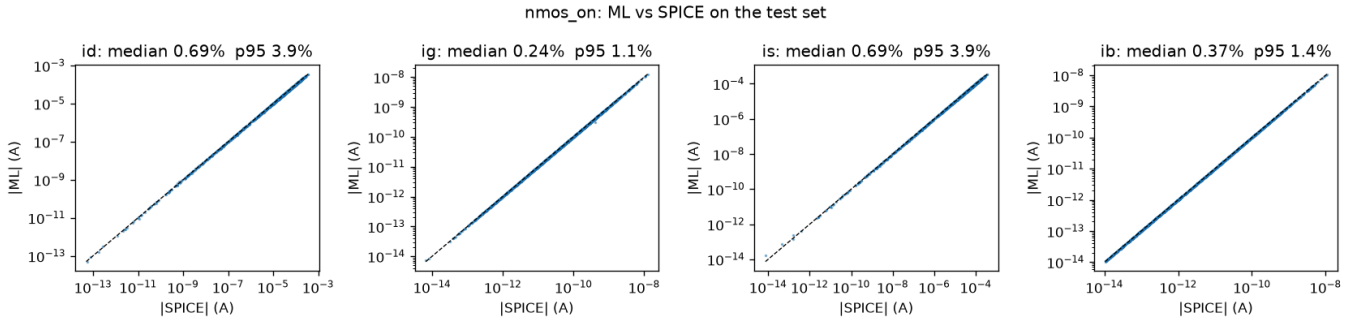


Figure 3. Predicted vs SPICE, NMOS ON, test set (Id, Ig, Is, Ib).

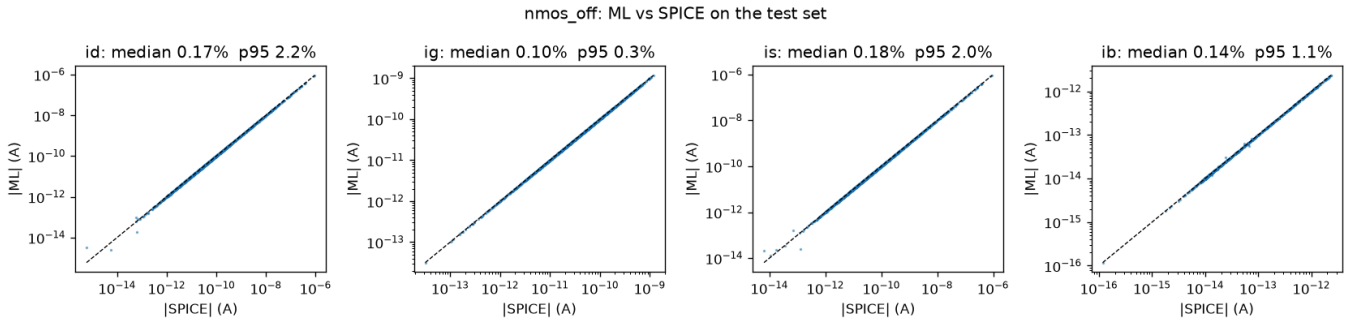


Figure 4. Predicted vs SPICE, NMOS OFF, test set.

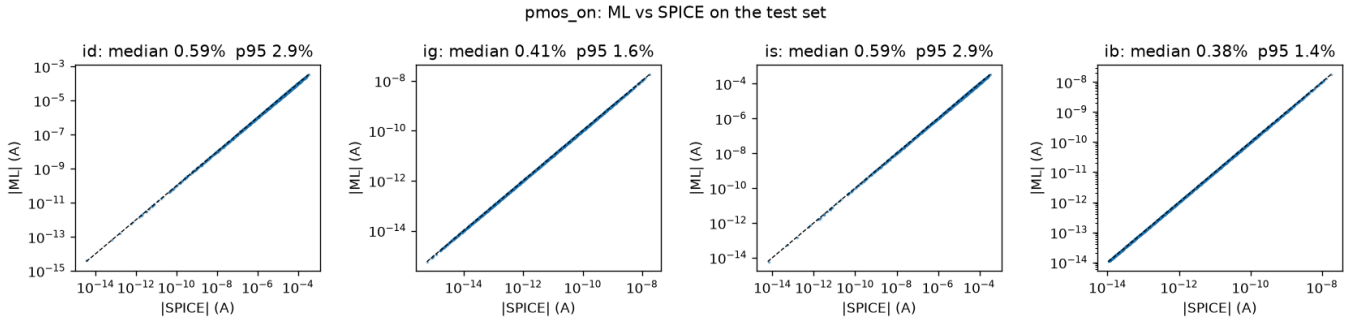


Figure 5. Predicted vs SPICE, PMOS ON, test set.

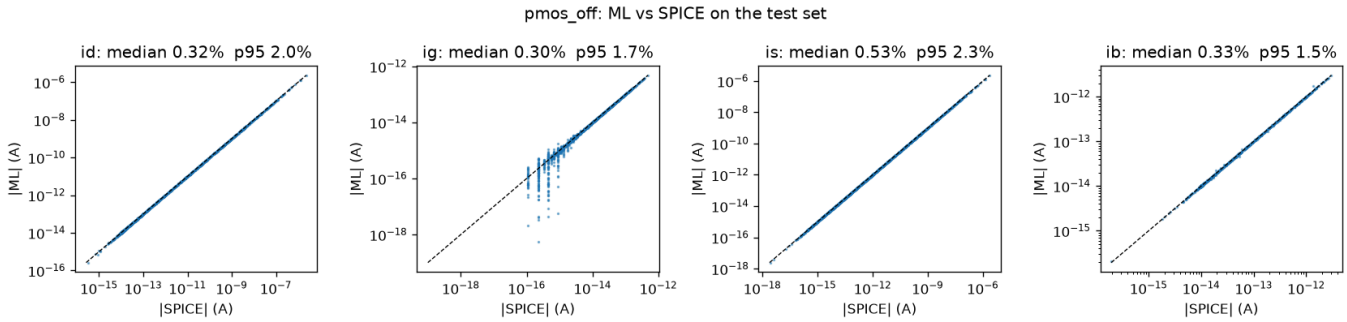


Figure 6. Predicted vs SPICE, PMOS OFF, test set. The spread in I_g is at 10^{-16}–10^{-15} A, below the evaluation floor.

3.4 Does the structure matter? Ablation and data size

The ablation trains the same MLP (same size, same rows, same ensemble) to predict each terminal current directly, on an asinh scale. It is 24.6× worse in median error (median over the 16 currents). The learning curve retrains the chosen recipe on 500 to 7000 uniform samples (plus half as many rail samples). The median error reaches within 25 % of the full-size model at NMOS ON n = 2000, NMOS OFF n = 4000, PMOS ON n = 2000, PMOS OFF n = 4000. This answers the brief's question of the smallest sufficient dataset: a few thousand SPICE samples per case.

Table 5. Ablation: direct terminal-current MLP vs the bipole model (validation set).

Case	current	direct med %	direct p95 %	bipole med %	bipole p95 %
NMOS ON	id	5.54	37.04	0.79	3.73
NMOS ON	ig	8.05	48.10	0.25	1.17
NMOS ON	is	5.09	32.94	0.79	3.73
NMOS ON	ib	9.63	42.91	0.39	1.46
NMOS OFF	id	6.32	91.95	0.18	2.30
NMOS OFF	ig	3.05	13.89	0.09	0.31
NMOS OFF	is	6.41	97.44	0.18	2.07
NMOS OFF	ib	0.50	9.92	0.14	1.02
PMOS ON	id	5.78	30.71	0.64	2.82
PMOS ON	ig	10.56	44.60	0.43	1.53
PMOS ON	is	6.13	31.71	0.64	2.81
PMOS ON	ib	9.73	38.08	0.39	1.33
PMOS OFF	id	9.02	76.19	0.28	2.09
PMOS OFF	ig	6.19	23.03	0.28	1.49
PMOS OFF	is	12.41	88.13	0.51	2.33
PMOS OFF	ib	0.81	11.70	0.34	1.33

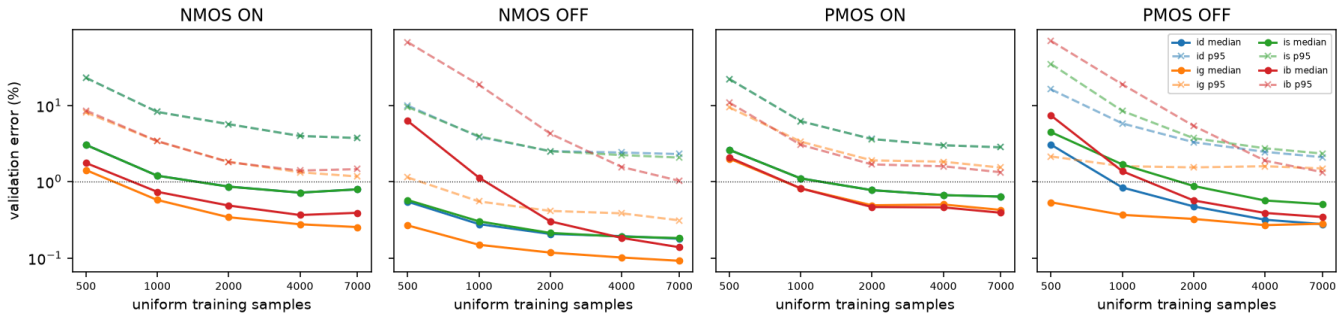


Figure 7. Learning curve: validation median (solid) and 95th-percentile (dashed) error vs the number of uniform training samples.

Table 6. Learning curve, validation median error (%).

Case	current	n = 500	n = 1000	n = 2000	n = 4000	n = 7000
NMOS OFF	ib	6.24	1.13	0.30	0.18	0.14
NMOS OFF	id	0.55	0.28	0.21	0.19	0.18
NMOS OFF	ig	0.27	0.15	0.12	0.10	0.09
NMOS OFF	is	0.57	0.30	0.21	0.19	0.18
NMOS ON	ib	1.76	0.73	0.49	0.36	0.39
NMOS ON	id	3.03	1.19	0.86	0.72	0.79
NMOS ON	ig	1.42	0.57	0.34	0.28	0.25
NMOS ON	is	3.03	1.19	0.86	0.72	0.79
PMOS OFF	ib	7.35	1.37	0.57	0.39	0.34
PMOS OFF	id	3.05	0.84	0.47	0.32	0.28
PMOS OFF	ig	0.54	0.37	0.32	0.27	0.28
PMOS OFF	is	4.45	1.68	0.88	0.57	0.51
PMOS ON	ib	2.07	0.82	0.46	0.46	0.39
PMOS ON	id	2.61	1.10	0.77	0.67	0.64
PMOS ON	ig	2.00	0.82	0.49	0.50	0.43
PMOS ON	is	2.61	1.10	0.77	0.67	0.64

4 Gate results

Each of the 18 gate/input cases is simulated at the same 200 random corners. A corner is one draw of VDD, T and process: one draw for all NMOS and one for all PMOS. Following the brief, the ML leakage uses the SPICE-recorded internal and output voltages to set each transistor's terminal voltages, then sums V·I over the predicted bipole currents. As an extension (§7.1), the internal and output node voltages are also solved from the ML models alone. Both sets of node voltages are recorded in `results/gate_samples.csv`.

Table 7. Gate results, 200 corners per case. Over all 3,600 simulations: mean accuracy 99.10 %, median error 0.69 %, p95 2.60 %, max 5.73 %. ML-solved: 99.12 %, 0.68 %, 2.42 %, 4.89 %.

Gate	inputs	SPICE median (A)	ML median (A)	bias %	median err %	p95 err %	max err %	ML-solved med %	ML-solved p95 %	max |ΔV| node (mV)
INV	0	1.61×10^{-7}	1.61×10^{-7}	-0.11	0.70	2.43	4.97	0.66	2.40	0.18
INV	1	8.08×10^{-7}	8.15×10^{-7}	+0.46	0.79	3.06	5.73	0.79	2.90	1.60
NAND2	00	7.92×10^{-10}	7.91×10^{-10}	-0.23	0.34	1.17	1.74	0.43	1.43	1.26
NAND2	01	1.57×10^{-7}	1.55×10^{-7}	-0.19	0.65	2.30	4.51	0.66	2.25	0.43
NAND2	10	1.72×10^{-8}	1.74×10^{-8}	+0.79	0.86	2.82	4.19	0.82	2.60	1.60
NAND2	11	1.34×10^{-6}	1.35×10^{-6}	+0.18	0.88	2.71	4.25	0.81	2.13	2.47
NOR2	00	3.18×10^{-7}	3.13×10^{-7}	-0.11	0.65	2.35	4.72	0.69	2.29	0.54
NOR2	01	6.48×10^{-7}	6.60×10^{-7}	+0.11	0.88	3.04	5.18	0.64	2.51	1.29
NOR2	10	4.92×10^{-8}	4.96×10^{-8}	+0.14	0.63	2.34	2.97	0.63	2.44	1.75
NOR2	11	9.47×10^{-10}	9.58×10^{-10}	+0.31	0.62	1.87	2.44	0.96	2.44	1.79
AND2	00	8.08×10^{-7}	8.16×10^{-7}	+0.46	0.79	3.06	5.73	0.79	2.90	1.60
AND2	01	1.04×10^{-6}	1.04×10^{-6}	+0.22	0.66	2.57	5.40	0.66	2.41	1.67
AND2	10	8.25×10^{-7}	8.34×10^{-7}	+0.44	0.75	3.01	5.71	0.77	2.84	1.62
AND2	11	1.70×10^{-6}	1.71×10^{-6}	+0.27	0.76	2.30	3.87	0.71	2.00	2.47
OR2	00	1.22×10^{-6}	1.23×10^{-6}	-0.01	0.59	2.32	5.08	0.58	2.16	1.57
OR2	01	8.83×10^{-7}	8.83×10^{-7}	+0.11	0.75	2.46	4.75	0.60	2.28	1.29
OR2	10	2.18×10^{-7}	2.18×10^{-7}	+0.00	0.61	2.19	4.42	0.58	2.14	1.75
OR2	11	1.62×10^{-7}	1.62×10^{-7}	-0.10	0.70	2.42	4.97	0.66	2.40	1.79

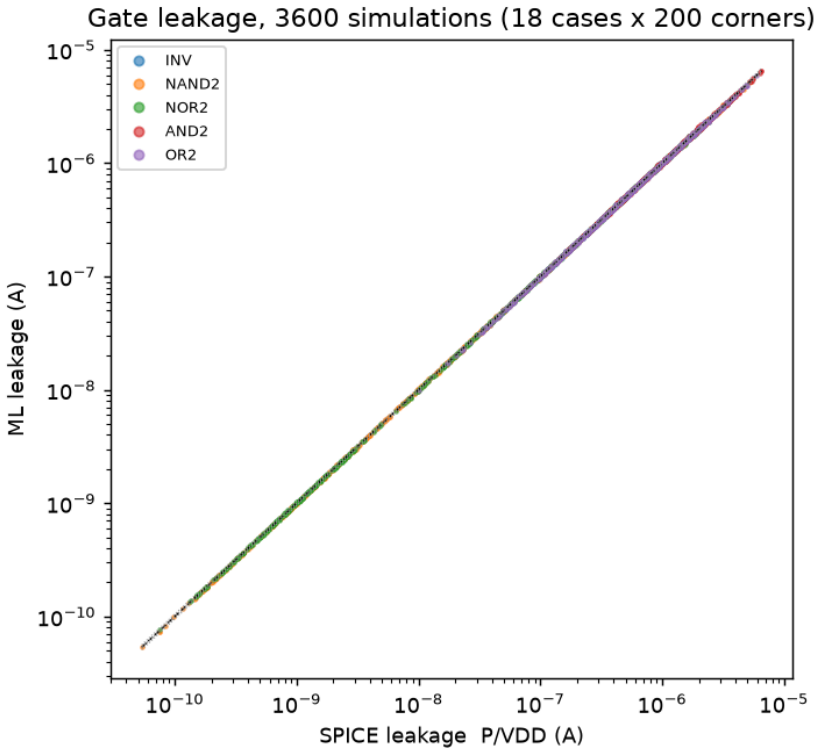


Figure 8. Gate parity plot, all 3,600 simulations (log axes; dotted lines at ±1 % and ±5 %).

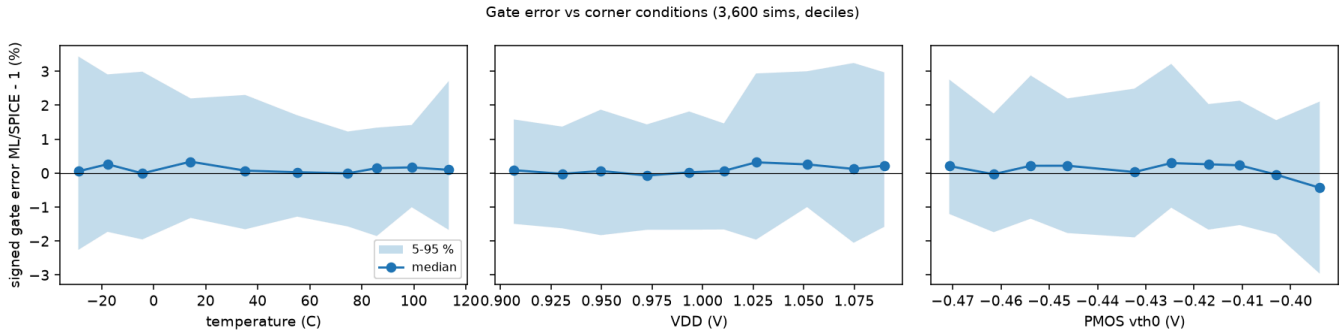


Figure 9. Signed gate error vs temperature, VDD and PMOS vth0, in deciles of each variable (median and 5–95 % band).

Most and least accurate. Mean accuracy per gate: INV 99.01 %, NAND2 99.15 %, NOR2 99.11 %, AND2 99.03 %, OR2 99.14 %. The most accurate case is NAND2 00 (p95 1.17 %). Its leakage is set by a two-transistor OFF stack, whose current is small and dominated by a single, well-sampled mechanism. The least accurate is INV 1 (p95 3.06 %). The cases with the largest tails (INV 1, AND2 00/01/10, whose output inverter sees a 1, and NOR2 01, OR2 00) all leak through a single OFF PMOS with $V_{sd} \approx$ VDD. That makes the PMOS-OFF channel model the accuracy bottleneck at gate level, consistent with it having the highest test p95 in Table 3. The stack effect is visible in the SPICE medians: NAND2 00 and NOR2 11 leak about 200× less than an inverter, because the internal node rises (falls) and reverse-biases the upper OFF device. The models reproduce this without being told: the ML-solved internal node voltages agree with SPICE to within 2.47 mV (median 0.051 mV).

5 Test-circuit results: 3-bit CLA

The adder is built only from library gates. The propagate signals $p_i = a_i \oplus b_i$ use 4-NAND2 XORs and the generate signals $g_i = a_i b_i$ use AND2. Each carry is computed directly from g, p and c_0 ($c_3 = g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_0$ from AND2/OR2 trees), and $s_i = p_i \oplus c_i$. That gives 42 gates and 204 transistors; a unit test checks that it adds correctly for all 128 inputs. SPICE simulates the full netlist for all 128 vectors at the same 200 corners (25,600 operating points, 112 s on 11 cores). For the hierarchical ML prediction (the brief's method), logic values are propagated, each gate's Part-B ML leakage is looked up for its input combination and corner, and the 42 values are summed.

Table 8. CLA metrics over 128 vectors × 200 corners = 25,600 points. Rows 2–4 split the error into its causes; row 5 is our extension (§7.2).

Comparison	SPICE median (A)	ML median (A)	bias %	median err %	p95 err %	max err %	mean acc %
hierarchical ML (gate sum) vs SPICE circuit	3.16×10^{-5}	3.07×10^{-5}	-2.56	2.59	10.41	21.20	96.41
SPICE gate sum vs SPICE circuit (hierarchy error)	3.16×10^{-5}	3.07×10^{-5}	-2.64	2.64	10.04	20.79	96.42
hierarchical ML (gate sum) vs SPICE gate sum (ML error)	3.07×10^{-5}	3.07×10^{-5}	+0.18	0.68	2.51	4.95	99.13
flat ML on circuit voltages vs SPICE circuit	3.16×10^{-5}	3.16×10^{-5}	+0.14	0.68	2.46	4.85	99.14
levels-aware ML hierarchy vs SPICE circuit (SPICE-free)	3.16×10^{-5}	3.17×10^{-5}	+0.28	0.66	2.18	3.52	99.21

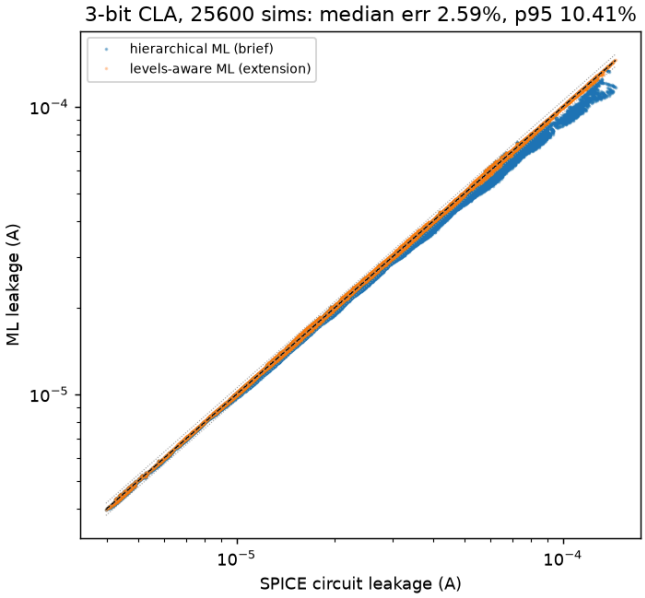


Figure 10. CLA parity plot (log axes): hierarchical ML (the brief) and levels-aware ML vs the SPICE circuit.

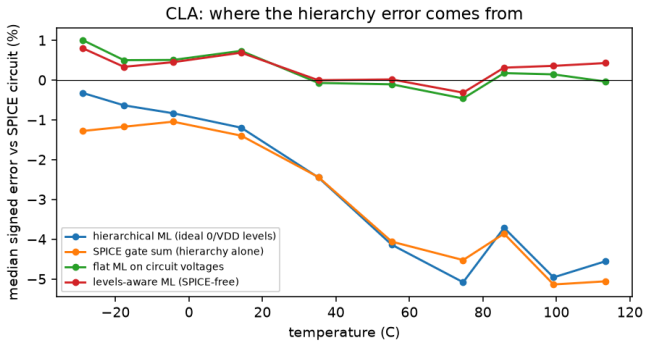


Figure 11. Median signed CLA error vs temperature for each method.

Compared with the gates. The hierarchical estimate reaches 96.41 % mean accuracy (median error 2.59 %, bias -2.56 %), clearly below the gate level, as the brief anticipates. The extra reference points show where the error comes from:

- **ML error alone** (ML gate sum vs SPICE gate sum): median 0.68 %, mean accuracy 99.13 %. This is the gate-level accuracy again; gate errors partly cancel in the sum.
- **Hierarchy error alone** (SPICE gate sum vs SPICE circuit): bias -2.64 %, p95 10.04 %. The isolated-gate assumption is wrong in a systematic way, independent of the ML.
- **Flat ML** (all 204 transistors at the circuit's own SPICE voltages): median 0.68 %. The transistor models are just as good inside the large circuit.

The hierarchy error grows with temperature, from about -1.1 % below 0 °C to -4.7 % above 80 °C. In a real chain a logic 0 sits tens of mV above ground (Figure 12) and a logic 1 below VDD, and the shift grows with leakage, so with temperature. The next gate's OFF transistor sees that offset on its gate, and sub-threshold current is exponential in it, so the real circuit leaks more than the ideal-input sum. Loading (tunnelling current drawn by fan-out pins) is a much smaller effect: the levels-aware hierarchy below ignores it and still removes the bias. Per vector, the median error ranges from 1.40 % (inputs 0001111) to 3.71 % (inputs 0110111); `results/circuit_results.csv` lists all 128.

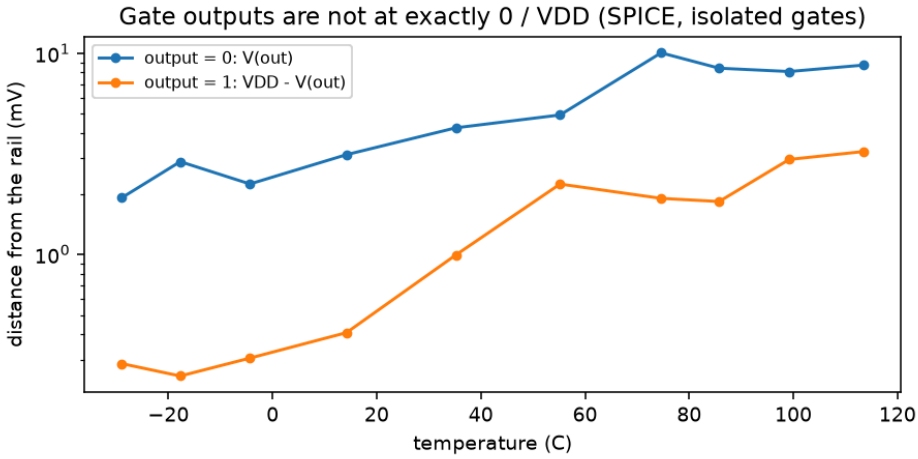


Figure 12. How far isolated-gate outputs sit from the rails in SPICE (median per temperature decile, log scale).

6 Error analysis

Where. At gate level the error depends mainly on temperature and supply. Below 0 °C the median |error| is 1.03 % (p95 3.18 %), against 0.57 % (p95 2.05 %) above 40 °C. Above 1.05 V the p95 is 3.17 %, against 2.04 % below 0.95 V. The worst points all come from a few cold, high-VDD corners (Table 9) and from the cases that leak through one OFF PMOS (§4). Process parameters have weak correlation with the error ($|r| < 0.1$ for every parameter), so the models capture the process dependence well.

Table 9. The eight largest gate errors.

Gate	inputs	corner	T (°C)	VDD (V)	NMOS vth0	PMOS vth0	error %
INV	1	76	-24.0	1.089	0.497	-0.428	+5.73
AND2	00	76	-24.0	1.089	0.497	-0.428	+5.73
AND2	10	76	-24.0	1.089	0.497	-0.428	+5.71
AND2	01	76	-24.0	1.089	0.497	-0.428	+5.40
NOR2	01	76	-24.0	1.089	0.497	-0.428	+5.18
OR2	00	76	-24.0	1.089	0.497	-0.428	+5.08
INV	0	184	-1.0	1.083	0.435	-0.420	+4.97
OR2	11	184	-1.0	1.083	0.435	-0.420	+4.97

Why. (1) Cold OFF currents are the smallest in the dataset, and log G changes fastest with V_{th}/V_T there. The same error in log G is a larger relative current error, and cold, high-VDD corners are corners of the 12-dimensional input box where uniform data is thinnest. (2) At high VDD, drain-induced barrier lowering and GIDL make the OFF current depend more steeply on V_{ds} exactly where the gate operates (V_{ds} = VDD). (3) The PMOS-OFF channel is the weakest of the four models, because its leakage depends on several competing mechanisms of similar size.

A SPICE artefact found along the way. Before the fix, the worst "error" was +126 % on INV A=1 at corner 198 (112 °C, 1.088 V). The ML was right: ngspice had converged to a spurious operating point. Sweeping V(out) with a forcing source shows a single true KCL root at 42 mV (3.58 μA). The default Newton start gets stuck at a narrow kink of the model near 22 mV. Starting every gate and circuit simulation from a logic-level `.nodeset` removes it. Only this one of the 3,600 gate points moved by more than 0.1 %, and no CLA point did.

Circuit level. The CLA error of the brief's method is dominated by the ideal-level assumption (§5), with a -2.64 % bias that grows with temperature. With the levels corrected, the remaining CLA error (0.66 % median) equals the gate-level ML error.

7 Extensions

7.1 SPICE-free node voltages

The ML models can predict the operating point as well as the leakage. At DC, the ML-predicted terminal currents into every internal node sum to zero. `leakml/solve.py` solves this KCL system for all unknown nodes at once: damped Newton, finite-difference Jacobian, each step limited to 50 mV as SPICE does. It is vectorised over corners and iterates only on rows that have not converged. The initial guess puts logic nodes at their rails and stack nodes at the rail of their ON neighbour. Every gate case converges, with the KCL residual below 10^{-12} A. The solved node voltages are within 2.47 mV of SPICE, and the leakage computed from them is as accurate as with SPICE voltages (99.12 % vs 99.10 %).

7.2 Levels-aware hierarchy

The same solver fixes the CLA hierarchy. Gates are visited in topological order, and each one is solved with its inputs set to the *ML-predicted output voltages* of its drivers instead of ideal 0/VDD. Rows that share a corner and input voltages are solved once. Over 42 gates the solver needs at most 12 iterations and leaves 0 rows unconverged. The predicted gate-output voltages match the full SPICE circuit to 0.044 mV in the median (2.90 mV max). The CLA leakage improves to **99.21 % mean accuracy**, with median error 0.66 %, p95 2.18 %, max

3.52 % and bias +0.28 %. That beats even the flat ML on SPICE voltages (99.14 %) and uses no SPICE input at all: the full chain from the model card to circuit leakage runs on four ML models.

7.3 Minimum-leakage standby vector

In standby, a block can be parked at the input vector that leaks least. The CLA's 128 vectors differ by 1.22× in mean leakage. Ranked by ML, the vectors follow SPICE with Spearman $\rho = 1.000$ (levels-aware) and 0.970 (ideal-level hierarchy). Both pick the same minimum-leakage vector as SPICE (1001000). Choosing the best vector separately in each corner, the levels-aware ML finds the exact SPICE optimum in 99 % of corners and is never more than 0.18 % above it.

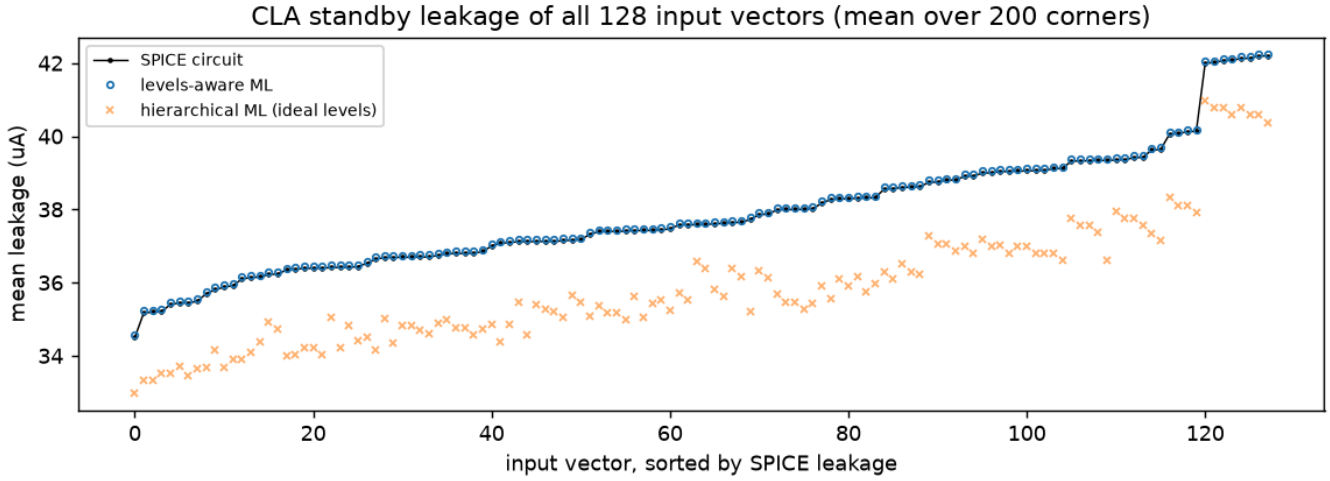


Figure 13. Mean CLA leakage of the 128 input vectors, sorted by SPICE.

7.4 Leakage by mechanism

SPICE reports only total currents, but the bipole models split leakage into physical mechanisms. Averaged over the 18 gate cases at nominal process and VDD:

- Channel (sub-threshold) leakage is 90.8 % at −40 °C, 93.3 % at 20 °C and 96.7 % at 125 °C.
- Gate tunnelling makes up the rest: 9.2 % when cold, falling to 3.3 % when hot. It is almost temperature-independent while sub-threshold current grows exponentially.
- Junction leakage and GIDL are below 0.016 % everywhere in this card.

Gate tunnelling matters most in the stacked cases, where the stack has suppressed sub-threshold leakage: it is 63 % of NAND2 00 and 55 % of NOR2 11 near room temperature.

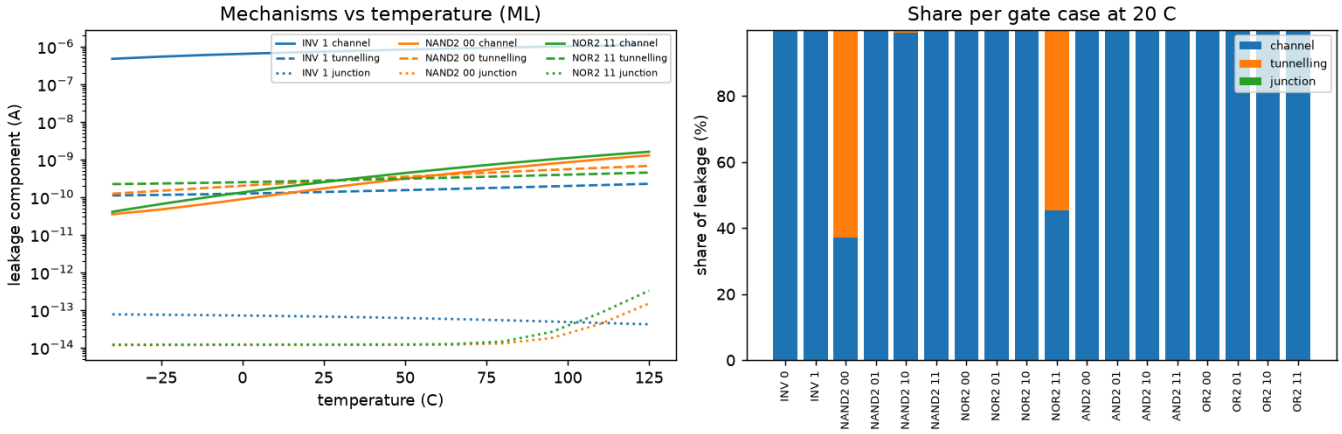


Figure 14. Left: mechanism currents vs temperature for three gate cases. Right: share of each mechanism per gate case near room temperature.

7.5 Sensitivity

Each PVT parameter is moved one at a time (process and VDD ±10 %, T to −40 / 125 °C) and the levels-aware CLA leakage is averaged over all 128 vectors. The largest effects are VDD -10% (0.37×), VDD +10% (2.61×), pmos vth0 +10% (0.44×), pmos vth0 -10% (2.18×). Supply voltage has the largest effect, through drain-induced barrier lowering: a higher V_{ds} lowers the OFF device's threshold. Next come PMOS vth0 and temperature, as the sub-threshold law $\exp(-V_{th}/nV_T)$ predicts. PMOS parameters matter more than NMOS ones because the PMOS devices are twice as wide and carry most of the CLA leakage. Oxide thickness, mobility and junction depth change the leakage by less than 10 %.

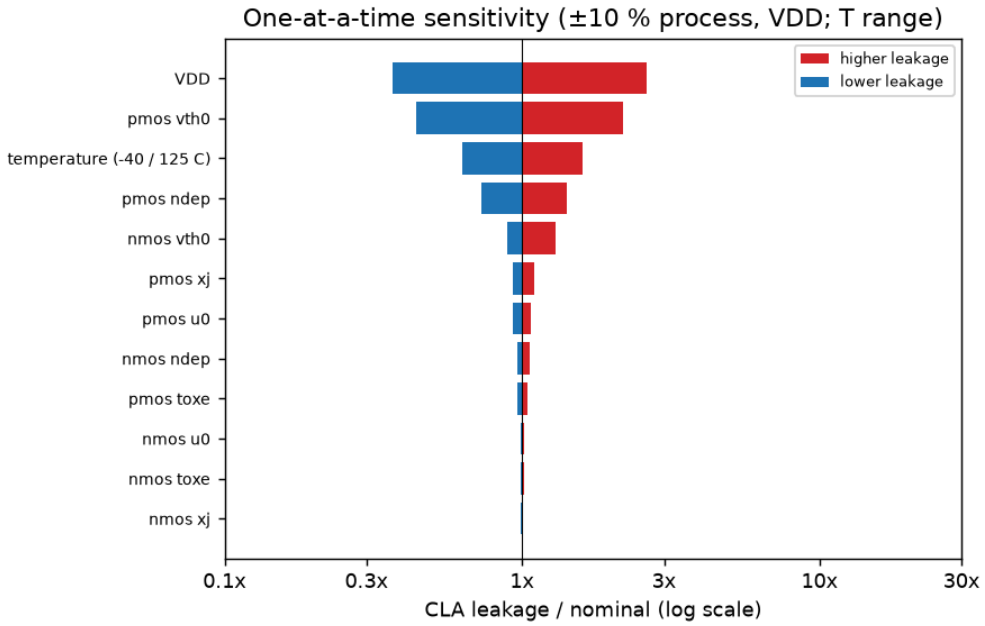


Figure 15. One-at-a-time sensitivity of the CLA leakage (log scale).

7.6 Monte-Carlo leakage distribution

Once trained, the models make large statistical studies cheap. The levels-aware CLA leakage is evaluated at 5,000 new random corners for the minimum- and maximum-leakage vectors. For vector 1001000, the median is 2.83×10^{-5} A and the 99th percentile 4.38× the median (lognormal $\sigma = 0.79$; the 200 SPICE corners give median 2.95×10^{-5} A, KS $p = 0.93$). For vector 1110001, the median is 3.50×10^{-5} A and the 99th percentile 4.19× the median (lognormal $\sigma = 0.77$; the 200 SPICE corners give median 3.64×10^{-5} A, KS $p = 0.89$). The distribution is strongly right-skewed: worst-case design must budget for several times the median leakage.

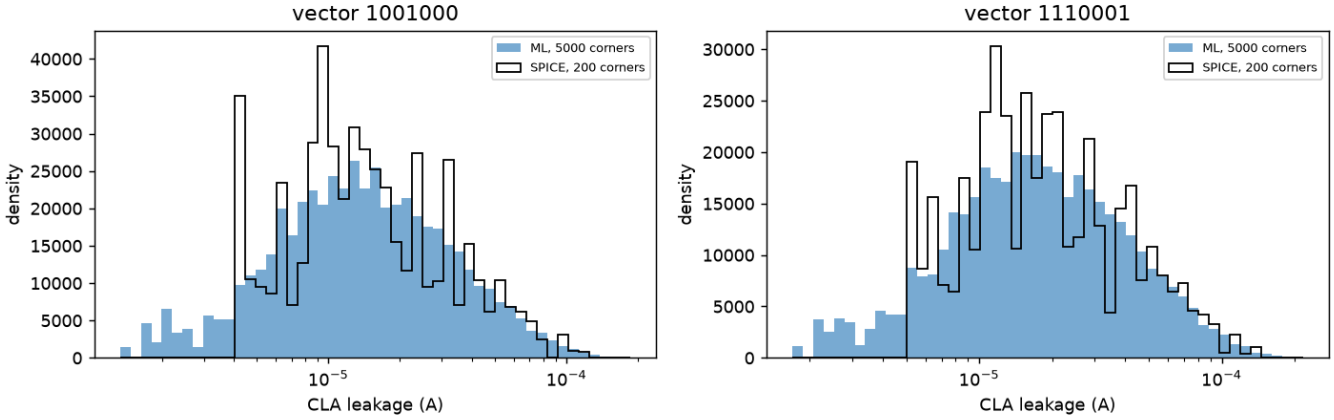


Figure 16. CLA leakage over new random corners (ML) vs the 200 SPICE corners.

7.7 Uncertainty from the ensemble

Each of the five ensemble members gives its own gate leakage, and their spread is a free error bar. The spread correlates only weakly with the real error (Spearman $\rho = 0.17$). The 20 % of points with the smallest spread have median error 0.60 %, and the 20 % with the largest have 0.79 %. The members disagree where data is thin, but most of the remaining error is shared by all members, a systematic bias of the training distribution. An ensemble cannot detect such a bias; only more data where the gates operate can remove it.

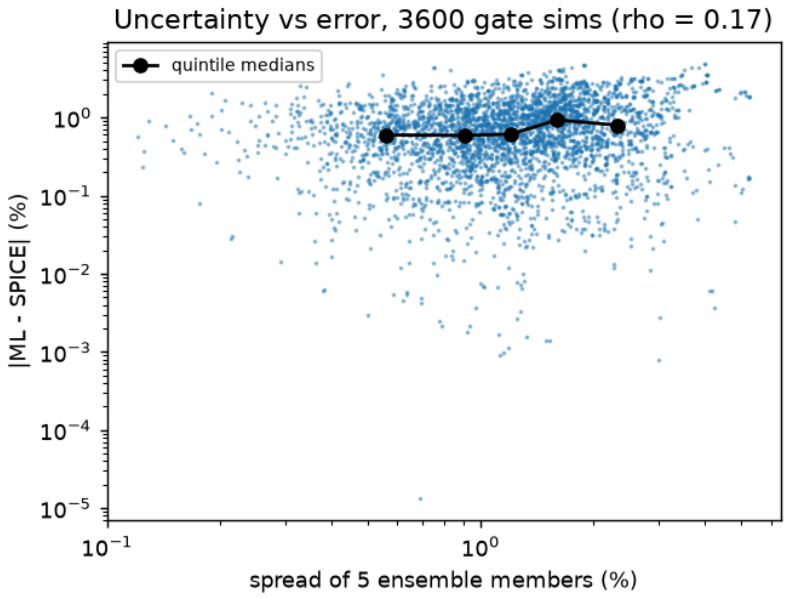


Figure 17. Ensemble spread vs actual error, all 3,600 gate simulations.

7.8 Cost

SPICE needs 48.07 ms of CPU time per CLA operating point. ML inference on the same 204 transistors, given the node voltages, needs 4.257 ms, 11× less. The brief's hierarchical lookup costs 1.7 s for all 25,600 points once the gate tables exist. The SPICE-free levels-aware solve of all 25,600 points takes 402 s on one core, about 3× less CPU time than SPICE. Most of that time goes into the finite-difference Jacobian, which needs many model calls per Newton step; an analytic Jacobian of the MLPs would remove most of it.

7.9 A second, unseen circuit: 8-bit parity checker

To check that nothing was tuned to the CLA, the brief's other test circuit was built from the same library: a balanced tree of seven 4-NAND2 XORs (28 gates, 112 transistors). It was simulated in SPICE for all 256 vectors at the same 200 corners (51,200 points), with no retraining or other change. The brief's hierarchy reaches 94.80 % mean accuracy (bias -4.04 %). The levels-aware ML hierarchy reaches **99.24 %** (median error 0.65 %, p95 1.96 %), confirming the CLA conclusions on a different circuit.

Table 10. 8-bit parity checker, 256 vectors × 200 corners.

Comparison	bias %	median err %	p95 err %	max err %	mean acc %
hierarchical ML (gate sum) vs SPICE circuit	-4.04	4.04	14.29	26.30	94.80
SPICE gate sum vs SPICE circuit (hierarchy error)	-4.08	4.08	13.82	25.94	94.74
hierarchical ML vs SPICE gate sum (ML error)	+0.14	0.71	2.35	4.15	99.12
levels-aware ML hierarchy vs SPICE circuit (SPICE-free)	+0.31	0.65	1.96	3.11	99.24

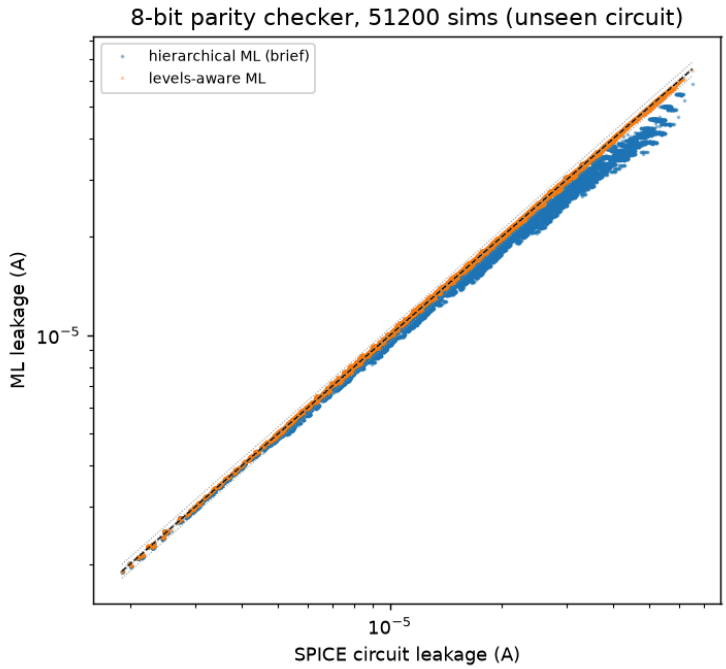


Figure 18. Parity checker: ML vs SPICE circuit leakage (log axes).

8 Limitations and improvements

- **Training distribution.** The 10,000 uniform samples rarely land where gates operate (terminals on the rails, cold and high-VDD corners). The rail supplement fixed most of this. The remaining tail is in cold, high-VDD, single-OFF-PMOS conditions. Adaptive sampling, adding SPICE points where the ensemble disagrees most, would target it directly.
- **Ideal-level hierarchy.** As specified in the brief, the hierarchy is biased by -2.64 % on this circuit whatever the ML: even exact SPICE gate leakages give only 96.42 % mean accuracy. The levels-aware hierarchy (§7.2) removes this limit. Loading between gates is still ignored, which is acceptable here because gate tunnelling is a small share of leakage at 16 nm HP (§7.4). It would matter more for thin-oxide LP cards.
- **Solver speed.** The levels-aware solve uses finite-difference Jacobians. An analytic Jacobian, or a small learned gate-output model, would make it as fast as the lookup.
- **Scope.** The models cover W = 1–8 L, ±10 % process and −40 to 125 °C. Outside that box they extrapolate. One process draw is shared by all devices of a type (no within-die mismatch), as specified. Mismatch would need per-transistor draws, which the models already support.
- **Simulator noise.** Data regenerated with ngspice 44.2 and 45.2 differs only below about 10^{-12} A. L-BFGS amplifies this into ±0.02 points of gate accuracy with 5 seeds (±0.07 with 3). Training itself is bit-reproducible on identical data.

9 Reproducibility and deliverables

All scripts run in order (`./run_all.sh --full`) from the model file with fixed seeds. The fast path (`./run_all.sh`) reuses the stored data, the saved models and the cached CLA SPICE reference, and reruns validation in minutes. `pytest` checks 26 invariants: KCL of the saved models, passivity of the bipoles, netlist/ML topology identity, truth tables, that the CLA adds correctly, the BSIM4 width clamp, solver convergence, and that the parity checker computes parity.

#	Deliverable	Where
1	Datasets (4 × 10,000 rows)	data/{nmos,pmos}_{on,off}.csv
2	Example netlists	netlists/single_*.cir, gate_*.cir, cla3.cir
3	Kirchhoff check, trend plots	results/kirchhoff_check.csv, figures/data_check_trends.png (Table 1, Figure 1)
4	Model comparison	results/model_comparison.csv (Table 2)
5	Chosen models	results/chosen_models_test.csv, figures/pred_vs_spice_*.png, saved_models/*.joblib (Table 3)
6	Gate results	results/gate_results.csv, results/gate_samples.csv (Table 7)
7	Gate parity plot	figures/gate_parity.png
8	CLA results	results/circuit_summary.csv, circuit_results.csv, figures/circuit_parity.png (Table 8)
9	Code	00_ … 08_*.py, leakml/, run_all.sh, tests/
10	Report	report/Leaky_Lima_Report.pdf (this document, generated by report/build_report.py)